Definir seed

set.seed(3702478)

```
sam <- rnorm(n = 10, mean = 1000, sd = 100)
```

```
> sam[1:10]
 [1]  789.5944 1006.5671 1115.2109  949.0632  980.2662 1018.9997 1062.7493
 [8]  981.8859 1068.4159 1048.3565
```

```
> mean(sam)
[1] 1002.111
```

```
> var(sam)
[1] 8019.143
```

```
# Histograma
ggplot(data = data.frame(sam)) +
+     geom_histogram(aes(x = sam), bins = 20, color = 'white', fill = "#10342d") +
+     labs(x = "Seed yield (kg/ha)", y = "Count") +
+     theme_minimal() +
+     theme(text = element_text(size = 18))
```

```
> # gráfico de densidade
> ggplot(data = data.frame(sam)) +
+     geom_density(aes(x = sam), color = 'white', fill = '#10342d', alpha = .75) +
+     labs(x = "Seed yield (kg/ha)", y = "Density") +
+     theme_minimal() +
+     theme(text = element_text(size = 18))
```

```
 # Boxplot
> ggplot(data = data.frame(sam)) +
+     geom_boxplot(aes(x = sam), fill = "#b2bc63") +
+     annotate(geom = 'point', x = mean(sam), y = 0, shape = 17, color = "#10342d", size = 3) +
+     theme_minimal() +
+     theme(axis.text.y = element_blank(), text = element_text(size = 18)) +
+     labs(x = "Seed yield (kg/ha)")
```

n = 100

```
> sam <- rnorm(n = 100, mean = 1000, sd = 100)
```

```
> sam[1:10]
 [1] 1007.0215  982.6667 1066.2597  879.8512 1001.9950 1007.4251  711.4177
 [8] 1071.6866  995.7705  914.4491
```

mean(sam) [1] 1015.727

var(sam) [1] 10309.43

```
> # Histograma
> ggplot(data = data.frame(sam)) +
+     geom_histogram(aes(x = sam), bins = 20, color = 'white', fill = "#10342d") +
+     labs(x = "Seed yield (kg/ha)", y = "Count") +
+     theme_minimal() +
+     theme(text = element_text(size = 18))
```

gráfico de densidade

ggplot(data = data.frame(sam)) + + geom_density(aes(x = sam), color = 'white', fill = '#10342d', alpha = .75) + + labs(x = "Seed yield (kg/ha)", y = "Density") + + theme_minimal() + + theme(text = element_text(size = 18))

Boxplot

ggplot(data = data.frame(sam)) + + geom_boxplot(aes(x = sam), fill = "#b2bc63") + + annotate(geom = 'point', x = mean(sam), y = 0, shape = 17, color = "#10342d", size = 3) + + theme_minimal() + + theme(axis.text.y = element_blank(), text = element_text(size = 18)) + + labs(x = "Seed yield (kg/ha)")

```
# n = 1000
```

```
> sam <- rnorm(n = 1000, mean = 1000, sd = 100)
```

```
> sam[1:10]
 [1]  854.6031 1059.4633 1001.6158  911.8401 1121.6526  957.6495 1047.9095
 [8] 1007.9861  913.3606 1155.7900
```

```
mean(sam)> mean(sam)
[1] 999.2293
```

```
> var(sam)
[1] 9791.693
```

```
> # Histograma
> ggplot(data = data.frame(sam)) +
+     geom_histogram(aes(x = sam), bins = 20, color = 'white', fill = "#10342d") +
+     labs(x = "Seed yield (kg/ha)", y = "Count") +
+     theme_minimal() +
+     theme(text = element_text(size = 18))
```

```
> # gráfico de densidade
> ggplot(data = data.frame(sam)) +
+     geom_density(aes(x = sam), color = 'white', fill = '#10342d', alpha = .75) +
+     labs(x = "Seed yield (kg/ha)", y = "Density") +
+     theme_minimal() +
+     theme(text = element_text(size = 18))
```

```
> # Boxplot
> ggplot(data = data.frame(sam)) +
+     geom_boxplot(aes(x = sam), fill = "#b2bc63") +
+     annotate(geom = 'point', x = mean(sam), y = 0, shape = 17, color = "#10342d", size = 3) +
+     theme_minimal() +
+     theme(axis.text.y = element_blank(), text = element_text(size = 18)) +
+     labs(x = "Seed yield (kg/ha)")
```